
Normal-Incidence Plane-Wave Reflection from a Dielectric Slab

Permittivity Sweep, Impedance Transformation, and CST
Validation

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Independent Engineering Portfolio Project

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Portfolio Context and Evidence

Project Context. This work originated from a graduate-level electromagnetic simulation exercise. The present document is an independently prepared public portfolio edition intended to demonstrate plane-wave boundary modeling, dielectric-slab impedance transformation, frequency-domain simulation, parameter-sweep interpretation, and engineering verification.

Public Portfolio Edition. The original academic material has been reorganized into the same restrained university-style architecture used by the other public engineering reports in this portfolio. Student-identification information, course and instructor details, institutional branding, and administrative submission language have been removed. The technical design inputs and preserved simulation evidence are retained. This document does not represent an official publication or endorsement by any university, instructor, employer, or software vendor.

Evidence Classification. Analytical evidence consists of the lossless transmission-line model for an air–dielectric–air section, the half-wavelength matching condition, and an independently executable Python script that regenerates the reflection magnitude and wrapped phase. Simulated evidence consists of preserved CST Studio Suite Learning Edition screenshots for geometry, boundary conditions, ports, parameter sweep, fundamental-mode reflection coefficient, and electric- and magnetic-field results. No fabricated hardware or laboratory measurement is involved.

Simulation Scope. The public source archive contains exported figures and LaTeX reports but not the native CST project, raw S-parameter exports, mesh statistics, or full solver logs. The numerical conclusions are therefore limited to the preserved screenshots and source-derived tables. The electric and magnetic side boundaries create a TEM-like unit-cell environment; they do not constitute an infinite free-space plane-wave model. A port-mode warning was reported during the sweep, and no formal mesh- or port-convergence study was preserved.

Prepared as part of the Hossein Molhem Engineering Portfolio.

Abstract

This report analyzes normal-incidence electromagnetic reflection from a finite, lossless dielectric slab placed between two air regions. The slab thickness is fixed at one free-space wavelength at 1 GHz, approximately 299.79 mm, while the relative permittivity is swept from 1 to 10 in increments of 0.25. The preserved CST frequency-domain results are interpreted using the equivalent transmission-line model of an air–dielectric–air section.

For an ideal slab of thickness $d = \lambda_0$, the half-wavelength matching condition predicts reflection minima at $\varepsilon_r = (m/2)^2$. Within the simulated range, the corresponding values are $\varepsilon_r = 1, 2.25, 4, 6.25$, and 9. The CST sweep exhibits minima at the same permittivity values. Preserved linear-magnitude values include $|S_{11}| = 1.94892 \times 10^{-4}$ at $\varepsilon_r = 1$, 3.21858×10^{-3} at 2.25, 1.04206×10^{-2} at 4, and 1.28988×10^{-2} at 6.25. These correspond to approximately $-74.20, -49.85, -39.64$, and -37.79 dB, respectively.

Electric- and magnetic-field plots at $\varepsilon_r = 4$ illustrate the TEM-like propagation environment and the matched electrical-length condition. The agreement in the locations of the reflection minima provides the principal validation result. Exact null depth is not treated as the primary metric because the CST model has finite transverse dimensions, numerical ports, finite mesh resolution, approximate side-boundary representation, and an unresolved higher-order port-mode warning.

Project Objective

The objective of this project is to establish a traceable analytical and numerical workflow for normal-incidence reflection from a finite dielectric slab. The study connects the intrinsic impedance and phase constant of a nonmagnetic dielectric to an equivalent transmission-line section and then tests the resulting matching condition using a CST parameter sweep.

The project has five principal technical objectives:

1. Derive the input impedance and reflection coefficient of an air–dielectric–air slab under normal incidence.
2. Show that a slab with physical thickness $d = \lambda_0$ becomes matched whenever its electrical thickness is an integer multiple of one-half wavelength, yielding $\varepsilon_r = (m/2)^2$.
3. Construct a two-port CST frequency-domain model with electric and magnetic side boundaries that approximates TEM-like propagation along the slab axis.
4. Sweep relative permittivity from 1 to 10 in 0.25 increments and compare the locations of the simulated $|S_{11}|$ minima with the analytical prediction.
5. Examine electric- and magnetic-field evidence at $\varepsilon_r = 4$, one of the predicted matched cases, while identifying the numerical limitations that constrain interpretation of the null depths and port-mode results.

The final outcome is a public engineering report consistent with the other portfolio projects, together with analytical verification code, machine-readable results, simulation evidence, and explicit claim boundaries.

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Chapter 1

Plane-Wave Reflection from a Dielectric Slab

1.1 Project Overview

A finite dielectric slab introduces two impedance discontinuities and a propagation phase delay. At normal incidence, the total reflected wave is the coherent sum of the reflection at the first interface and the multiple reflections that return from the second interface. The result depends on both relative permittivity and electrical thickness [1], [2].

This study uses a fixed physical slab thickness of approximately one free-space wavelength at 1 GHz and sweeps the dielectric relative permittivity. The analytical model predicts a repeated set of matched conditions because the wavelength inside the dielectric changes as $1/\sqrt{\epsilon_r}$. A CST frequency-domain model is used to test the predicted locations of the reflection minima.

Table 1.1: Principal analytical and simulated results.

Quantity	Result
Operating frequency	1.000 GHz
Free-space wavelength	299.792 mm analytically; 299.79 mm used in CST geometry
Slab thickness	$d \approx \lambda_0 = 299.79$ mm
Permittivity sweep	$\epsilon_r = 1$ to 10 in 0.25 increments, 37 points
Predicted matched values	$\epsilon_r = 1, 2.25, 4, 6.25, 9$
CST minimum locations	Same five sweep values
Representative CST minimum	$\epsilon_r = 4$: $ S_{11} = 0.0104206$ or -39.64 dB
Field-validation case	$\epsilon_r = 4$ at 1 GHz
Evidence status	Analytical and simulated; no measured validation

1.2 Engineering Scope and Evidence

The public project is defined by the fixed model specification in Table 1.2. The study evaluates the fundamental-mode input reflection coefficient at one frequency while varying material permittivity.

Table 1.2: Public model specification.

Category	Parameter	Specified value
Electrical	Evaluation frequency	1.000 GHz
Geometry	Slab thickness	299.79 mm, approximately λ_0
Geometry	Transverse dimensions	$H_0 = 44.87$ mm, $W_0 = 299.79$ mm
Material	Relative permittivity	1 to 10 in 0.25 increments
Material	Relative permeability	1
Material	Loss model	Lossless dielectric
Solver	Method	CST frequency-domain solver
Ports	Configuration	Two waveguide ports, three modes requested per port
Boundaries	Side faces	Electric on x faces, magnetic on y faces
Boundaries	Longitudinal faces	Open on z faces
Monitors	Field case	E and H fields at 1 GHz for $\epsilon_r = 4$

1.2.1 Evidence Classes

Analytical evidence consists of the transmission-line equations, the half-wavelength matching derivation, regenerated reflection curves, and executable Python verification.

Simulated evidence consists of the preserved CST screenshots. The parameter sweep, $|S_{11}|$, S_{11} phase, model geometry, boundaries, port definitions, and field plots are directly visible. Selected numerical null values are preserved in the source report and result-table screenshot.

Measured evidence is absent. The project contains no fabricated specimen, material measurement, field-probe scan, or calibrated microwave measurement.

1.2.2 Public Claim Boundary

The report claims that the observed locations of the simulated reflection minima agree with the analytical matching condition. It does not claim an exact infinite-plane-wave solution, convergence-qualified null depth, material-tolerance robustness, or measured performance. The fundamental-mode result is interpreted despite a reported port-mode accuracy warning because its trend follows the analytical prediction; this agreement does not replace a formal port- and mesh-convergence study.

1.3 Theory and Analytical Model

The modeled structure is an air–dielectric–air stack under normal incidence. The dielectric is lossless and nonmagnetic, so

$$\mu_r = 1, \quad \varepsilon = \varepsilon_r \varepsilon_0. \quad (1.1)$$

1.3.1 Wave Impedance and Phase Constant

The intrinsic impedance and phase constant inside the dielectric are

$$\eta_d = \sqrt{\frac{\mu_0}{\varepsilon_0 \varepsilon_r}} = \frac{\eta_0}{\sqrt{\varepsilon_r}}, \quad (1.2)$$

and

$$\beta_d = k_0 \sqrt{\varepsilon_r} = \frac{2\pi}{\lambda_0} \sqrt{\varepsilon_r}. \quad (1.3)$$

At 1 GHz,

$$\lambda_0 = \frac{c}{f_0} = 299.792 \text{ mm}. \quad (1.4)$$

1.3.2 Equivalent Transmission-Line Section

The slab is represented as a lossless transmission-line section of characteristic impedance η_d , length d , and load impedance $Z_L = \eta_0$. The input impedance looking into the slab is

$$Z_{\text{in}} = \eta_d \frac{Z_L + j\eta_d \tan(\beta_d d)}{\eta_d + jZ_L \tan(\beta_d d)}. \quad (1.5)$$

The analytical input reflection coefficient is

$$\Gamma_{\text{ana}} = \frac{Z_{\text{in}} - \eta_0}{Z_{\text{in}} + \eta_0}. \quad (1.6)$$

1.3.3 Half-Wavelength Matching Condition

When the line length is an integer multiple of one-half wavelength in the dielectric, the input impedance repeats the load impedance:

$$d = m \frac{\lambda_d}{2}, \quad \lambda_d = \frac{\lambda_0}{\sqrt{\varepsilon_r}}. \quad (1.7)$$

For $d = \lambda_0$,

$$\lambda_0 = m \frac{\lambda_0}{2\sqrt{\varepsilon_r}}, \quad (1.8)$$

$$\sqrt{\varepsilon_r} = \frac{m}{2}, \quad (1.9)$$

$$\varepsilon_r = \left(\frac{m}{2}\right)^2. \quad (1.10)$$

Table 1.3: Analytically predicted matched permittivity values for $d = \lambda_0$.

Half-wave index m	$\sqrt{\varepsilon_r} = m/2$	Predicted ε_r
2	1.0	1.00
3	1.5	2.25
4	2.0	4.00
5	2.5	6.25
6	3.0	9.00

Figure 1.1 shows the regenerated ideal analytical response. The zeros occur at the values in Table 1.3. The analytical phase is mathematically ill-conditioned at an exact zero because a complex number at the origin has no unique phase.

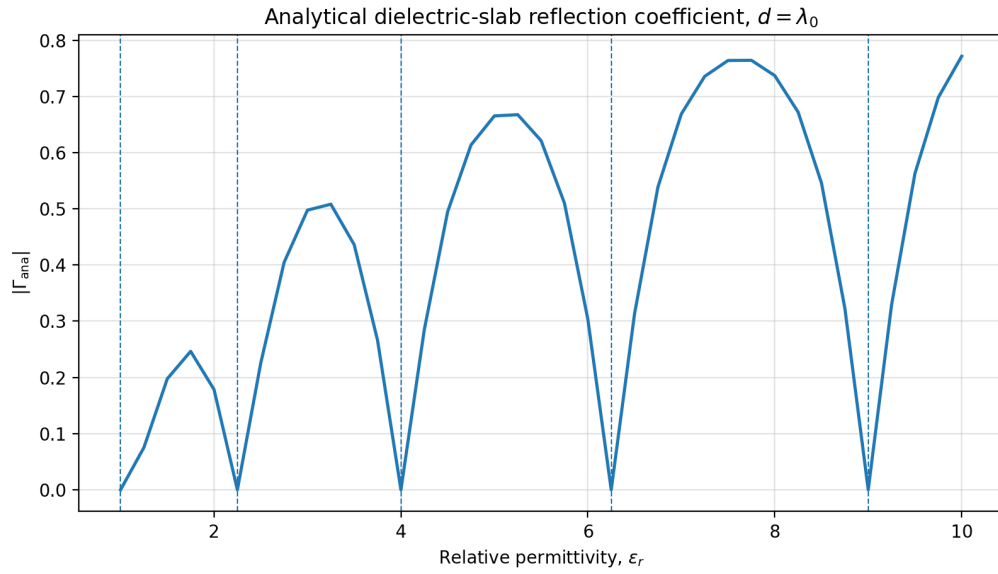


Figure 1.1: Regenerated analytical reflection magnitude for the ideal air–dielectric–air slab with $d = \lambda_0$.

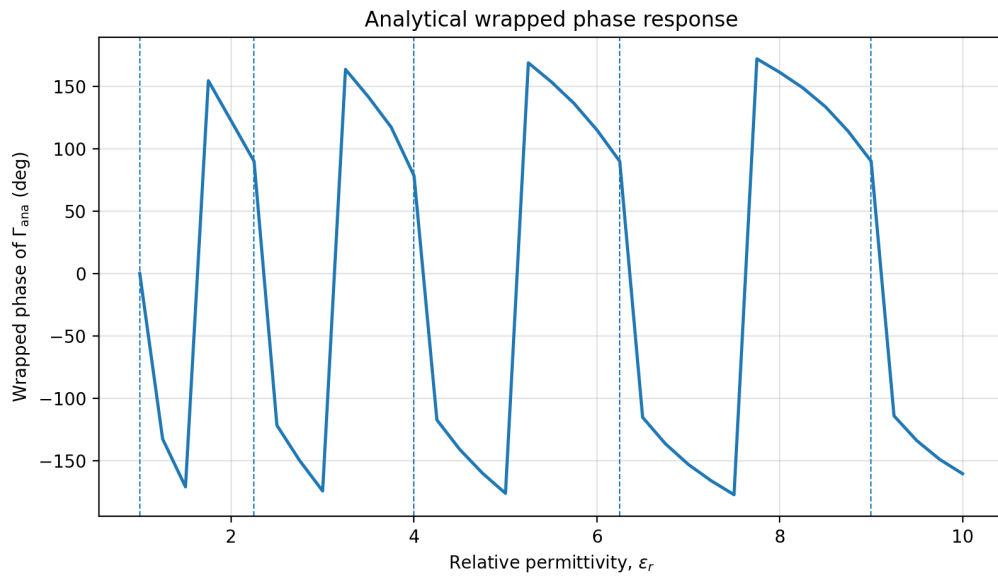


Figure 1.2: Regenerated wrapped analytical phase. Values at or extremely near the reflection zeros are numerically sensitive and are not used as matching metrics.

1.4 CST Model and Simulation Method

The CST model consists of an input air region, a parameterized dielectric slab, and an output air region. Geometry is defined in millimeters and frequency in GHz. The frequency-domain solver evaluates the structure up to 1 GHz, with electric- and magnetic-field monitors at 1 GHz.

1.4.1 Geometry and Material Parameterization

The principal dimensions are

$$H_0 = 44.87 \text{ mm}, \quad W_0 = 299.79 \text{ mm}, \quad L_0 = 299.79 \text{ mm}. \quad (1.11)$$

The slab thickness is L_0 , approximately one free-space wavelength at 1 GHz. The dielectric relative permittivity is controlled by the CST parameter `eps`; the outer regions are vacuum.

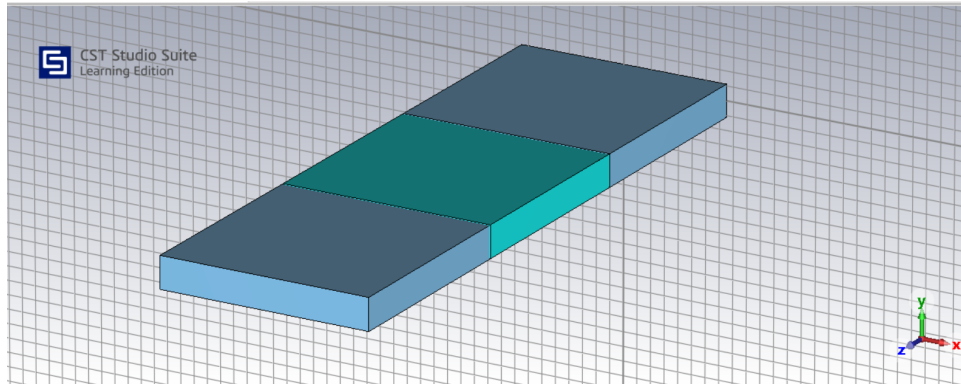


Figure 1.3: Preserved CST geometry of the air–dielectric–air structure. The center brick is the parameterized dielectric slab.

1.4.2 Boundary Conditions

Electric boundaries are applied to the two x -normal side faces and magnetic boundaries to the two y -normal side faces. The two longitudinal z faces are open. Together, the paired electric and magnetic walls create a TEM-like propagation environment in the finite transverse model.

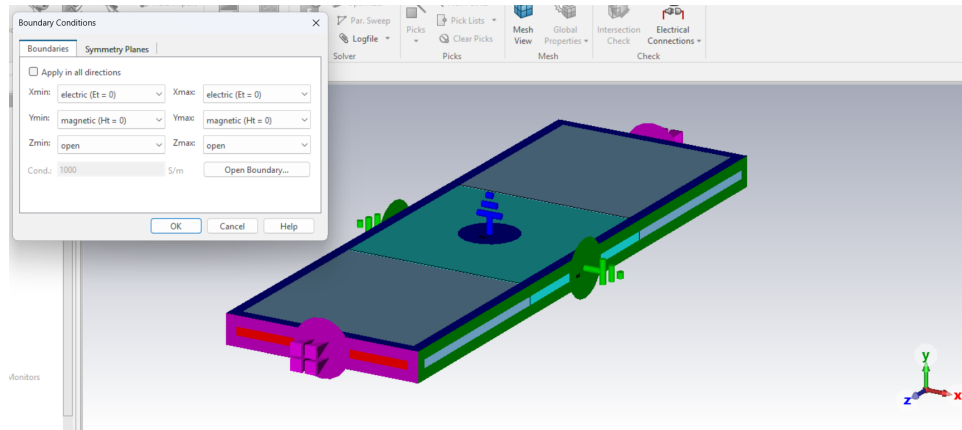
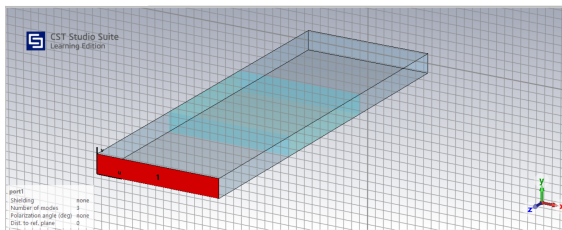


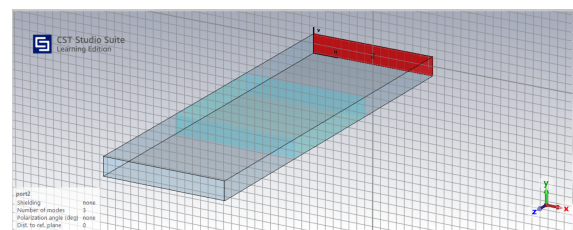
Figure 1.4: Preserved CST boundary-condition definition.

1.4.3 Ports and Modes

Two waveguide ports are placed at the longitudinal ends of the model. The source report states that one propagating mode was initially insufficient and that three modes were requested at each port. The primary metric remains the fundamental-mode reflection coefficient $S_{1(1),1(1)}$, denoted S_{11} in the public report.



(a) Port 1.



(b) Port 2.

Figure 1.5: Preserved CST waveguide-port definitions.

1.4.4 Permittivity Sweep

The parameter ϵ_{ps} is swept linearly from 1 to 10 using 37 points, corresponding to a 0.25 increment. The S_{11} magnitude and phase are extracted at 1 GHz.

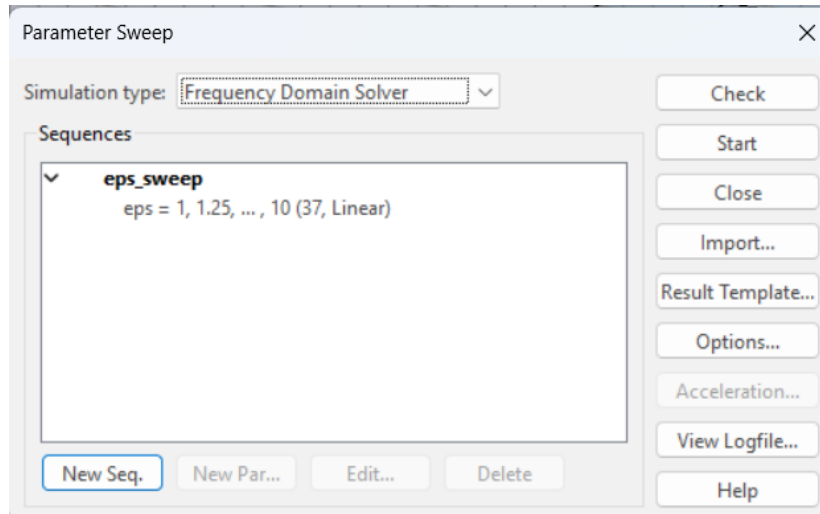


Figure 1.6: CST parameter-sweep definition for relative permittivity.

1.5 Results and Verification

1.5.1 Reflection Magnitude

Figure 1.7 shows the preserved CST fundamental-mode reflection magnitude at 1 GHz. The curve exhibits minima at $\varepsilon_r = 1, 2.25, 4, 6.25,$ and 9 , matching the analytical sequence from Eq. (1.10).

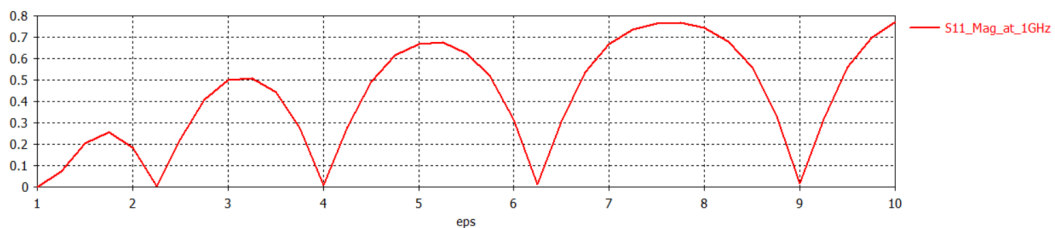
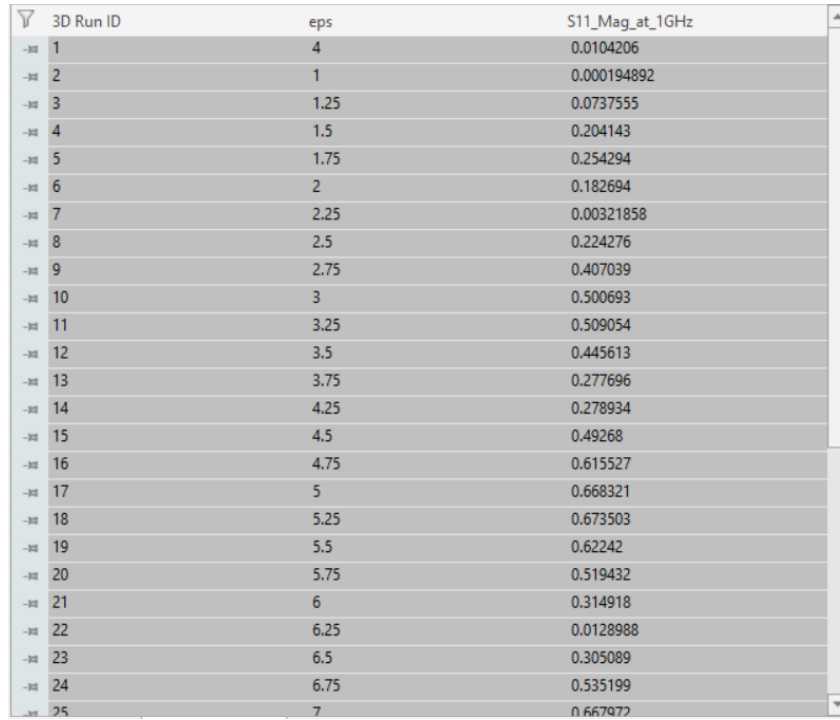


Figure 1.7: Preserved CST $|S_{11}|$ versus relative permittivity at 1 GHz.

Table 1.4: Selected CST reflection minima preserved in the source report.

ε_r	CST $ S_{11} $	$20 \log_{10} S_{11} $	Evidence note
1.00	0.000194892	-74.20 dB	Preserved sweep value
2.25	0.00321858	-49.85 dB	Preserved sweep value
4.00	0.0104206	-39.64 dB	Preserved initial validation value
6.25	0.0128988	-37.79 dB	Preserved sweep value
9.00	Near zero	Not numerically preserved	Deep minimum visible in sweep plot

The source archive also contains the CST result-table screenshot shown in Fig. 1.8. It provides traceability for the parameterized sweep but does not replace a raw exported data file.



3D Run ID	eps	S11_Mag_at_1GHz
1	4	0.0104206
2	1	0.000194892
3	1.25	0.0737555
4	1.5	0.204143
5	1.75	0.254294
6	2	0.182694
7	2.25	0.00321858
8	2.5	0.224276
9	2.75	0.407039
10	3	0.500693
11	3.25	0.509054
12	3.5	0.445613
13	3.75	0.277696
14	4.25	0.278934
15	4.5	0.49268
16	4.75	0.615527
17	5	0.668321
18	5.25	0.673503
19	5.5	0.62242
20	5.75	0.519432
21	6	0.314918
22	6.25	0.0128988
23	6.5	0.305089
24	6.75	0.535199
25	7	0.667972

Figure 1.8: Preserved CST result table for selected $|S_{11}|$ sweep values.

1.5.2 Reflection Phase

The preserved phase response is shown in Fig. 1.9. Rapid transitions and wrapping occur near the magnitude minima. This behavior is expected because phase becomes highly sensitive when the complex reflection coefficient approaches the origin.

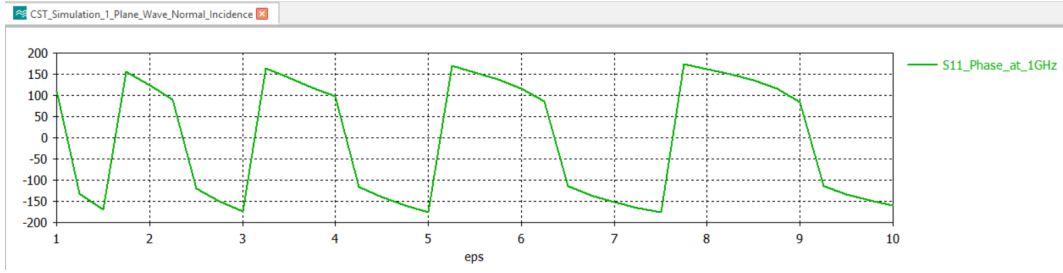


Figure 1.9: Preserved CST phase of S_{11} at 1 GHz.

1.5.3 Validation Metric

The primary validation metric is the location of each minimum, not its exact depth. All five analytically predicted matched permittivity values occur on the 0.25 sweep grid and are visible as minima in the CST response. Table 1.5 summarizes this comparison.

Table 1.5: Analytical-to-CST comparison of reflection-minimum locations.

m	Analytical ε_r	CST minimum location	Location difference
2	1.00	1.00	0.00
3	2.25	2.25	0.00
4	4.00	4.00	0.00
5	6.25	6.25	0.00
6	9.00	9.00	0.00

The zero location differences reflect the fact that the analytical matched values coincide exactly with sampled sweep points. They do not establish that the CST null depths have converged to the ideal analytical zeros.

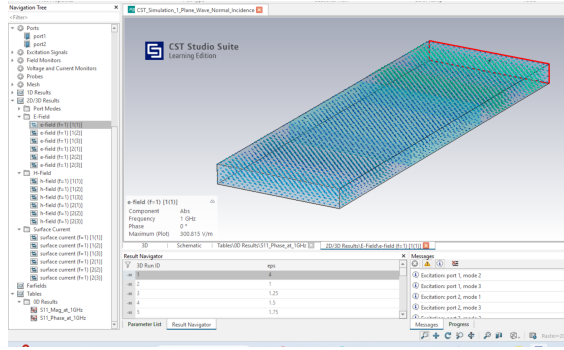
1.6 Field Evidence and Engineering Discussion

1.6.1 Matched Field Case at $\varepsilon_r = 4$

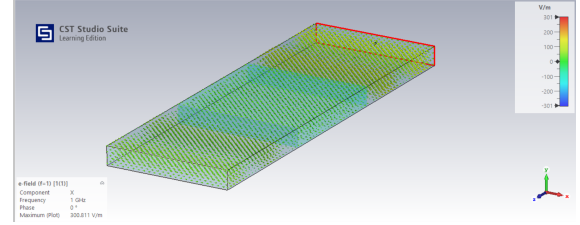
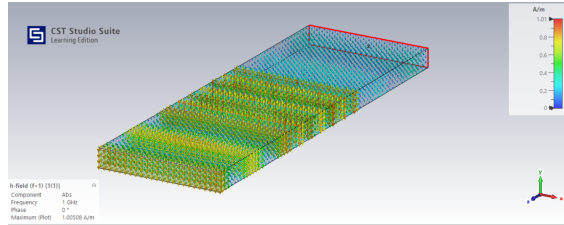
At $\varepsilon_r = 4$,

$$\lambda_d = \frac{\lambda_0}{2}, \quad d = \lambda_0 = 2\lambda_d = 4\frac{\lambda_d}{2}. \quad (1.12)$$

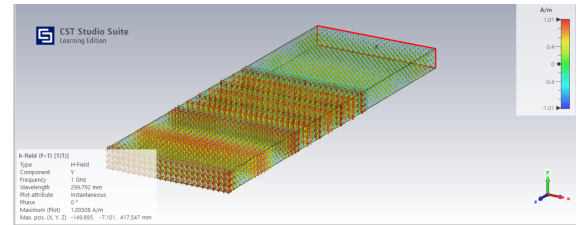
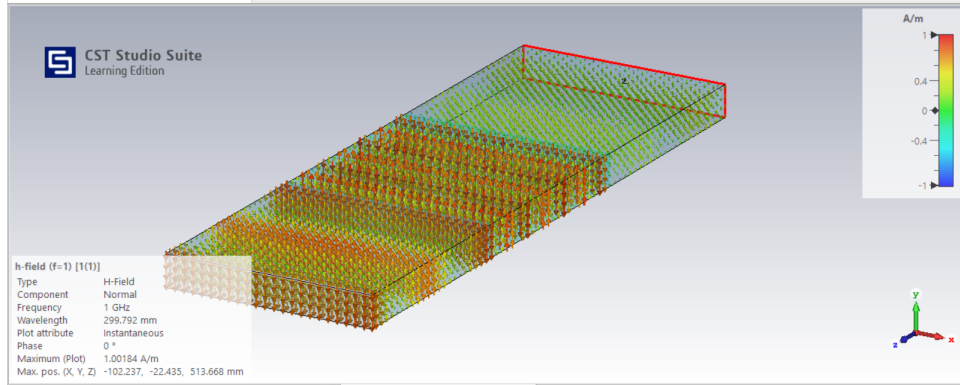
The slab therefore contains four half wavelengths and transforms the output-air impedance back to the input plane. The preserved field plots are evaluated at this matched sweep value.



(a) Electric-field magnitude.

(b) Dominant x -directed electric field.**Figure 1.10:** Preserved electric-field evidence at 1 GHz for $\epsilon_r = 4$.

(a) Magnetic-field magnitude.

(b) Dominant y -directed magnetic field.**Figure 1.11:** Preserved magnetic-field evidence at 1 GHz for $\epsilon_r = 4$.**Figure 1.12:** Preserved normal magnetic-field component for the selected observation surface at $\epsilon_r = 4$.

The dominant transverse electric and magnetic components are consistent with the intended TEM-like propagation environment. The screenshots are qualitative field evidence; no raw field data or quantitative field-uniformity metric is available.

1.6.2 Physical Interpretation of the Minima

Each interface produces a partial reflection because the intrinsic impedance changes from η_0 to η_d and back to η_0 . The input reflection is the coherent sum of these contributions after repeated internal propagation. At an integer half-wavelength electrical thickness, the input impedance repeats the air load, so the net reflection approaches zero.

1.6.3 Why the CST Nulls Are Nonzero

The ideal analytical model assumes infinite transverse extent, perfect normal incidence, an exactly lossless homogeneous slab, and exact numerical evaluation. The preserved CST model instead uses finite transverse dimensions, waveguide ports, finite mesh resolution, side-wall boundary approximations, and a rounded physical length. These differences make small nonzero minima expected.

1.6.4 Port-Mode Warning

The source report records a warning that some modes at Port 2 did not reach good accuracy and that a non-standard mode was detected. Agreement of the fundamental-mode minimum locations with theory supports the intended physical interpretation but does not prove that higher-order modal content is negligible. A stronger study would refine the port mesh, vary the number of requested modes, inspect modal fields and cutoff behavior, and demonstrate convergence of $S_{1(1),1(1)}$.

1.6.5 Numerical Reliability

No mesh-convergence, boundary-size sensitivity, port-length sensitivity, or raw-data comparison is preserved. Consequently, the report treats the minimum locations as validated qualitative and discrete quantitative evidence, while treating the exact null depths as fixed-model outputs rather than convergence-qualified predictions.

1.7 Engineering Verification Matrix

Table 1.6 maps the principal engineering claims to the evidence retained in the public package.

Table 1.6: Engineering verification matrix.

Claim	Evidence	Outcome	Status	Next validation action
Half-wave slab matching occurs at $\varepsilon_r = (m/2)^2$	Transmission-line derivation and executable analytical script	1, 2.25, 4, 6.25, 9	Analytically verified	Extend to lossy and oblique-incidence cases
CST minimum locations follow the analytical sequence	Preserved $ S_{11} $ sweep and source result table	All five minima occur at the predicted sweep points	Supported by preserved simulation evidence	Export raw complex S-parameter data
The $\varepsilon_r = 4$ case satisfies an integer half-wave condition	Analytical wave-length relation and field screenshots	$d = 2\lambda_d = 4(\lambda_d/2)$	Analytically verified and qualitatively supported	Quantify phase progression from raw field data
The model supports TEM-like propagation	Side-boundary setup and dominant E_x , H_y screenshots	Dominant transverse components visible	Supported qualitatively	Calculate modal purity and cross-component ratios
The exact CST null depths represent the ideal plane-wave solution	No convergence study; finite model and port warning	Nulls are small but nonzero	Not established	Complete mesh, boundary, port-mode, and length convergence
The project provides measured dielectric-slab validation	No fabricated sample or measurement record	No measured evidence	Not claimed	Fabricate a slab and measure free-space or guided reflection

1.8 Conclusion

A lossless air–dielectric–air slab was analyzed using both an equivalent transmission-line model and a CST frequency-domain simulation. The slab thickness was approximately one free-space wavelength at 1 GHz, and relative permittivity was swept from 1 to 10 in 0.25 increments.

The analytical model predicts matched conditions at $\varepsilon_r = (m/2)^2$. The five values

within the sweep range are 1, 2.25, 4, 6.25, and 9. The preserved CST $|S_{11}|$ response contains minima at the same five values, which is the central validation result. Field plots at $\varepsilon_r = 4$ are consistent with the intended transverse propagation environment and the integer half-wavelength condition.

The report does not interpret the exact null depths as convergence-qualified values. Finite transverse geometry, approximate side boundaries, numerical ports, finite mesh, rounded slab length, and a recorded port-mode warning distinguish the CST model from the ideal infinite plane-wave problem. Future work should preserve the native model and raw complex data, establish mesh and port convergence, examine modal purity, include loss and oblique incidence, and perform experimental reflection measurements.

Appendix A

Analytical Verification

This appendix provides the executable Python script used to regenerate the ideal lossless reflection magnitude and wrapped phase over the documented permittivity sweep.

A.1 Verification Script

```

1  """Analytical verification for normal incidence on a lossless dielectric slab.
2
3  The model is air -> dielectric slab -> air at 1 GHz. The ideal analytical
4  slab thickness is one exact free-space wavelength. The script regenerates
5  reflection magnitude and wrapped phase over eps_r = 1...10 in 0.25 steps,
6  prints the predicted matching points, and writes analytical figures and CSV.
7  """
8  from __future__ import annotations
9
10 from pathlib import Path
11 import csv
12 import math
13
14 import matplotlib.pyplot as plt
15 import numpy as np
16
17 C0 = 299_792_458.0
18 ETA0 = 376.730313668
19 FO_HZ = 1.0e9
20 LAMBDAO_M = C0 / FO_HZ
21 D_M = LAMBDAO_M
22 EPS_VALUES = np.arange(1.0, 10.0 + 0.25, 0.25)
23
24
25 def slab_reflection(eps_r: float) -> complex:
26     """Return the ideal input reflection coefficient of an air-slab-air section."""
27     if eps_r <= 0:

```

```

28     raise ValueError("eps_r must be positive")
29     eta_d = ETA0 / math.sqrt(eps_r)
30     beta_d = 2.0 * math.pi * math.sqrt(eps_r) / LAMBDAA0_M
31     tangent = math.tan(beta_d * D_M)
32     z_in = eta_d * (ETA0 + 1j * eta_d * tangent) / (
33         eta_d + 1j * ETA0 * tangent
34     )
35     return (z_in - ETA0) / (z_in + ETA0)
36
37
38 def main() -> None:
39     out_dir = Path(__file__).resolve().parent.parent / "figures" / "analytical"
40     out_dir.mkdir(parents=True, exist_ok=True)
41     csv_path = Path(__file__).resolve().parent.parent / "results" /
42         "analytical-sweep.csv"
43
44     gammas = np.array([slab_reflection(float(eps)) for eps in EPS_VALUES])
45     magnitudes = np.abs(gammas)
46     phases_deg = np.angle(gammas, deg=True)
47
48     predicted = [(m, (m / 2.0) ** 2) for m in range(2, 7)]
49
50     print("Normal-incidence dielectric-slab analytical verification")
51     print(f"f0 = {F0_HZ / 1e9:.6f} GHz")
52     print(f"lambda0 = {LAMBDAA0_M * 1e3:.6f} mm")
53     print(f"ideal slab thickness d = lambda0 = {D_M * 1e3:.6f} mm")
54     print(f"eta0 = {ETA0:.6f} ohm")
55     print("sweep: eps_r = 1.00 to 10.00 in 0.25 steps")
56     print()
57     print("Predicted half-wave matching points:")
58     for m, eps_r in predicted:
59         gamma = slab_reflection(eps_r)
60         print(
61             f"    m={m}: eps_r={eps_r:.2f}, |Gamma|={abs(gamma):.6e}, "
62             f"    phase={math.degrees(math.atan2(gamma.imag, gamma.real)):.3f} deg"
63         )
64
65     with csv_path.open("w", newline="", encoding="utf-8") as stream:
66         writer = csv.writer(stream)
67         writer.writerow(["eps_r", "gamma_real", "gamma_imag", "gamma_mag",
68             "phase_deg"])
69         for eps_r, gamma, mag, phase in zip(EPS_VALUES, gammas, magnitudes,
70             phases_deg):
71             writer.writerow(
72                 [
73                     f"{eps_r:.2f}",
74                     f"{gamma.real:.12e}",
75                     f"{gamma.imag:.12e}",
76                     f"{mag:.12e}",
77                     f"{phase:.9f}",
78                 ]
79             )

```



```

77     plt.figure(figsize=(8.2, 4.8))
78     plt.plot(EPS_VALUES, magnitudes, linewidth=1.8)
79     for _, eps_r in predicted:
80         plt.axvline(eps_r, linestyle="--", linewidth=0.8)
81     plt.xlabel(r"Relative permittivity, $\varepsilon_r$")
82     plt.ylabel(r"$|\Gamma_{\mathrm{ana}}|$")
83     plt.title(r"Analytical dielectric-slab reflection coefficient, $d=\lambda_0$")
84     plt.grid(True, alpha=0.35)
85     plt.tight_layout()
86     plt.savefig(out_dir / "analytical_gamma_vs_eps.png", dpi=200)
87     plt.close()
88
89     plt.figure(figsize=(8.2, 4.8))
90     plt.plot(EPS_VALUES, phases_deg, linewidth=1.8)
91     for _, eps_r in predicted:
92         plt.axvline(eps_r, linestyle="--", linewidth=0.8)
93     plt.xlabel(r"Relative permittivity, $\varepsilon_r$")
94     plt.ylabel(r"Wrapped phase of $\Gamma_{\mathrm{ana}}$ (deg)")
95     plt.title(r"Analytical wrapped phase response")
96     plt.grid(True, alpha=0.35)
97     plt.tight_layout()
98     plt.savefig(out_dir / "analytical_phase_vs_eps.png", dpi=200)
99     plt.close()
100
101     print()
102     print(f"Wrote analytical sweep: {csv_path}")
103     print(f"Wrote figures: {out_dir}")
104
105
106
107 if __name__ == "__main__":
108     main()

```

Listing A.1: Python verification of the dielectric-slab reflection response.

A.2 Console Output

```

Normal-incidence dielectric-slab analytical verification
f0 = 1.000000 GHz
lambda0 = 299.792458 mm
ideal slab thickness d = lambda0 = 299.792458 mm
eta0 = 376.730314 ohm
sweep: eps_r = 1.00 to 10.00 in 0.25 steps

Predicted half-wave matching points:
m=2: eps_r=1.00, |Gamma|=7.544312e-17, phase=0.000 deg
m=3: eps_r=2.25, |Gamma|=1.530808e-16, phase=90.000 deg
m=4: eps_r=4.00, |Gamma|=3.750601e-16, phase=78.396 deg
m=5: eps_r=6.25, |Gamma|=2.508114e-15, phase=90.000 deg

```

```
m=6: eps_r=9.00, |Gamma|=9.797174e-16, phase=90.000 deg  
Wrote analytical sweep: /mnt/data/plane-wave-dielectric-slab-portfolio-package/  
    results/analytical-sweep.csv  
Wrote figures: /mnt/data/plane-wave-dielectric-slab-portfolio-package/figures/  
    analytical
```

Listing A.2: Captured output from the analytical verification script.

Appendix B

Simulation Evidence Record

B.1 Run Manifest

Table B.1: Preserved CST evidence record.

Record	Frequency or value	Evidence
Geometry	$d = 299.79$ mm	CST geometry screenshot
Boundary setup	Electric x , magnetic y , open z	CST boundary screenshot
Ports	Two ports, three requested modes each	Port screenshots and source report
Sweep	$\varepsilon_r = 1$ to 10, step 0.25	Parameter-sweep screenshot
Reflection magnitude	1 GHz	CST curve and result table
Reflection phase	1 GHz	CST phase curve
Field case	$\varepsilon_r = 4$, 1 GHz	E- and H-field screenshots

B.2 Evidence Availability

The public package includes the screenshots used in the report, the LaTeX source, analytical verification code, regenerated analytical figures, and machine-readable result summaries. The native CST project, raw complex sweep data, mesh statistics, solver log, port-mode field exports, and formal convergence records are not contained in the supplied archive and are therefore not claimed as reproducibility assets.

References

- [1] C. A. Balanis, *Advanced Engineering Electromagnetics*, 2nd ed. John Wiley & Sons, 2012.
- [2] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. John Wiley & Sons, 2013.